Physics
Higher level
Markscheme
B.3 Gas laws

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W4ss Academy Paper 2 practice 2 | **Topic** B.3 Gas laws | **Ref.** W4SS–PHY–B3–P2–02

Student name

Paper 2

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- Award one mark per ✓. Accept equivalent answers (OWTTE) and error carried forward (ECF) where stated.
- Total: **[20 marks]**.

Paper 2 — markscheme

Question	Marks	Answers
1. (a)(i)	**[1]**	Mass-energy «of uranium» into kinetic energy of fission products ✓
1. (a)(ii)	**[3]**	Mass of uranium nucleus ≈ 235u ✓ $\frac{\text{energy}}{\text{mass}} = \frac{180 \times 10^{6} \times 1.60 \times 10^{-19}}{235 \times 1.661 \times 10^{-27}}$ ✓ 7.4×10^{13} «J» ✓
1. (b)(i)	**[2]**	beta minus decay ✓ $^{132}_{52}$Te has more neutrons / higher $\frac{N}{Z}$ ratio than stable nuclides of similar A «and beta minus reduces $\frac{N}{Z}$ » ✓
1. (b)(ii)	**[1]**	$e^{25} = 7.2 \times 10^{10}$ «s^{-1}» ✓
1. (b)(iii)	**[1]**	Takes ln of both sides of $A = A_0 e^{-\lambda t}$, leading to $\ln A = \ln A_0 - \lambda t$ ✓ «hence slope $= -\lambda$ » ✓
1. (b)(iv)	**[2]**	Slope =«−» 2.5×10^{-6} «s^{-1}» ✓ $T_{\frac{1}{2}} = \text{«} \frac{\ln 2}{2.5 \times 10^{-6} \times 60 \times 60 \times 24} = \text{» 3.2 «days»}$ ✓
1. (c)(i)	**[1]**	Collects thermal energy from the coolant and delivers it to the gas ✓ Prevents the «irradiated» coolant from leaving the reactor vessel ✓
1. (c)(ii)	**[3]**	Correct read offs of pressure and volume at B ✓ $8.0 \times 10^{6} \times 1.6 \times 10^{-3} = 1.0 \times 8.31 \times T$ ✓ T = 1500 «K» ✓
1. (c)(iii)	**[1]**	From $\Delta S = \frac{\Delta Q}{T}$, the change in entropy is zero when $\Delta Q = 0$ ✓
1. (c)(iv)	**[2]**	Net work done = « 8.23 + 9.11 - 4.32 - 3.25 » 9.77 «kJ» ✓ Efficiency =« $\frac{9.77}{20.58}$ = » 0.47 ✓
1. (d)	**[3]**	Rotational KE is proportional to ω^2 ✓ Calculation of ω^2 for at least four points, e.g. {96.1, 76.7, 57.6, 38.4, 19.3}×10^{3} Shows that the differences in equal time intervals are approximately the same, e.g. {19.4, 19.1, 19.2, 19.1, 19.3}×10^{3} ✓ *Allow a tolerance of ±1×10^{3} s^{-2} from the values stated in MP2.*